Extending Mental Poker

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Abstract. Mental Poker enables two players to play card games at a distance without having a physical deck of cards at hand, as long as they have reliable trapdoor permutations or access to Oblivious Transfer (OT). OT itself can be stored and extended using just a one-way function. We examine whether Mental Poker can be extended from initial hands of ordinary physical poker.

On the theoretical side, this work establishes OT amplification/extraction as a cryptographic primitive using asymmetrically-viewed fully-randomizing permutations. On the pragmatic, it is naturally related to "card cryptography" although in a very restricted sense: 52-card deck protocols using fully scrambling shuffles. While secret-key exchange protocols make strong use of full shuffles, full shuffles are avoided in 2PC/AND/Solitaire card settings because of detrimental over-randomization. Within those domains, this work explains decades-old impossibility conjectures in certain circumstances. More positively, we provide several practical and information-theoretically secure ways to establish base OTs from 5-Card Draw over ordinary decks at rates sufficient to establish a foothold for indefinite Mental Poker extension in just an afternoon of play.

1 Introduction

We address an elementary question that has hidden in plain sight for decades:

Is Poker complete for Mental Poker? (And for cryptography in general?)

After her acrimonious telephonic divorce, Alice admitted to Eve over coffee that she and Bob still occasionally play phone games now and then. It turns out they'd played a few hands on vacation back in the day, and somehow now, when it fits their current whims and idle relationship statuses, they pick up a hand or two of Mental Poker. They'll always have Paris, Alice joked.

It seems plausible, to Eve. After all, Eve was thoroughly deceived by their private-key exchange protocols over Bridge many times over [14], back when she was just the friendly nosy neighbor of the ill-fated couple. But Alice did seem a bit too insistent when arguing that when she has the Queen of Hearts, Bob can't possibly have her. "It. I mean. Not 'her,' " Alice corrected.

Asymmetric knowledge is fair in love and war, and poker, after all. Why shouldn't asymmetrically-correlated poker hands trivially give rise to games, to OT, to extended OT, and to unlimited MPC?

Well. Let's rewind the story a bit to help Eve out.

1.1 Beginnings

A half century ago, Shamir, Rivest, and Adleman [33] proposed the game of Mental Poker, played using cryptographic tools over the phone. Mental Poker became the canonical example of secure general multiparty computation of functions more varied than poker, inspiring a series of foundational works [5,8,15,39] and a decades-long line of fruitful research and application.

In a celebrated result, Kilian [22] showed that Multiparty Computation (MPC) can be powered by Oblivious Transfer (OT). Later authors showed OT and hence MPC can be battery-powered using rechargeable batteries: namely, preprocessed and extended [3, 4, 19] using just a one-way function.

Oblivious Transfer [31] enables Bob to learn one of two secrets held by Alice without Alice discovering which he learned. (There are variants, immaterial right now.) If you have a trusted party, or a network of helpers, or a one-way trapdoor permutation, you can generally implement OT, but at significantly more expense than symmetric-key operations.

The OT (and respectively, private-AND(x,y)) architecture is so simple and powerful that a significant line of research addresses how to purify, amplify and correct a wide variety of imperfect versions of these primitives [1, 7, 20, 28, 30, 37, 38].

While it's natural to explore gate-like primitives to build general computation, Mental Poker is hardly a gate-like primitive. Or is it? Ignoring most of the rules and interactions of poker, it's essentially a secret permutation viewed partly by Alice and partly by Bob. Theoretical and practical questions abound.

Can we extend Mental Poker?
Can we base OT on Mental Poker?
Are card-based protocols relevant?
How practical are any potential solutions?

In this work, we'll see some surprising limitations and some inspiring successes - including human-scale OT extraction from an afternoon's worth of poker hands. Our goal is to adhere to Five-Card Draw. The essence is to examine the power of a $(5, 5; 52)$ or a $(10, 10; 52)$ deal - namely, a couple of 5-card hands (or 10-cards, after drawing) from a freshly shuffled deck of 52 cards.

We restrict ourselves to a shuffling machine and a dealer, allowing Alice and Bob to converse, but otherwise foregoing the liberties of specialized cards or unusual decks. In this respect, our scenario is distinct from the broad design space of "card cryptography" [14, 26].

Thus we are in a kind of intersection: card crypto is relevant, but on a purely theoretical level this work focuses on full-range permutations arriving from "somewhere" (a dealer, a primitive, a commodity, a quantum transition, a prepared initialization).

Indeed, in contrast, card cryptography techniques for solving AND(x,y) generally employ cuts (random rotations) to randomize hidden sequences of cards, avoiding the sheer chaotic destruction of full shuffles. The decades-old *Solitary Games Conjecture* [11] predicts that fully-randomized shuffles interfere so strongly that card-based solitaire protocols cannot be built on them. We'll connect these issues to Generalized Erasure Channels (GEC), offering reasons why simply dealing full shuffles appears insufficient in such scenarios.

And one should note: the dealers in Vegas, Monte Carlo and Casablanca don't offer to limit themselves to cuts.

Surprisingly, the tables turn dramatically when jokers enter the scene. In other card games like Magic the Gathering, for example, pre-game selection of a deck to deploy is part of the fun and strategy. By cleverly building a 52-card deck from the collection of 54 cards in any toy store deck, we can use ordinary poker with a fully shuffling dealer to establish OT-extractable correlations. This is an extremely unexpected result given the longstanding Solitary Games Conjecture (see Conjecture 4).

As a slight diversion, we also explore how to morph card-based secret-key exchange protocols [14] into OT by marking the decks. On the faces, not the backs! Feeding front-marked "one-time decks" into a shuffling machine is quite effective for producing base OTs at low cost. This pedagogical parlor trick motivates understanding into the more pure approaches.

These techniques, while interesting in their own right, ultimately inspire a mark-free build-free protocol, establishing OT from untouched standard 52-decks using communication and rejection sampling, free of the pessimistic limitations of Solitary Games and stark Generalized Erasure Channels.

Altogether, we are able to convince Eve that Alice's questionable Unlimited Mental Poker claims aren't purely "Mental:"

- Full shuffles fail in specific scenarios, with GEC-related intuitions
- Marking decks help capitalize on classic key-exchange card crypto protocols
- Building 52-card decks from a 54-card library provides extractable OT
- Interactive rejection-sampling emulates deck-building to obtain OT by modifying raw GECs

In the last century, people met to exchange encryption keys in person so they could communicate securely from then on. In this century, people can meet with a deck of cards in person to establish unlimited secure mutual calculations into the future.[1]

1.2 Navigation

Sections 2 and 3 for background. Section 4 packs trios into decks to achieve significant OT rates. Section 5 explores marked decks. Section 6 explores building decks from a supply of ordinary cards - and specifically the use of jokers to unlock OT from 52/54-card off-the-shelf-decks. Section 8 explores limitations of full shuffling for unmarked decks. Section 7 emulates deck building using rejection sampling. Section 9 concludes.

[1] In fact, some did, in Paris, in July of 2025. They'll always have Paris, too.

2 Notation

We address poker-style protocols using full shuffles, allowing a simple definition of a "deal." (In a fully-general world with cuts of a deck implementing rotations of sequences, a deal might need to include the order in which cards are deployed and received, but we can avoid such complexity.)

A *deck*, for our purposes, is just a set of integers $\{0, ..., n-1\}$. For descriptive purposes, we may apply the syntactic sugar of French-style four suits *shdc* and thirteen labels *A23456789TJQK*, using terminology like *Ah* to denote the Ace of Hearts, also denoted as h_0. The Jack of Clubs: *Jc* or c_{10}.

Definition 1. *A* deal signature *[14] is a specification of hand sizes and draw pile size. For two parties, we denote a deal of* a *cards to Alice and* b *cards to Bob from a deck of* d *cards as* $(a, b; d)$.

Five-Card Draw is a Poker game using the standard 52-card deck and an initial deal signature of $(5, 5; 52)$. Abstracting away many of the easily-found details of the game, each player can opt to replace some or all of their hand by drawing further cards. The game can thus support a $(10, 10; 52)$ deal when Alice and Bob greedily replace all they can. It can support $(5, 10; 52)$ deals, etc., if one desires. It's not particularly challenging to define a n-card generalization $(m, m; n)$ and to perform asymptotic analyses, obtaining positive results, but we will focus our energies on $n = 52$.

Rabin Oblivious Transfer is a 50-50 noisy 1-bit channel via which Alice sends bit b to receiver Bob, who obtains with equal probability a success and data result, $(\top, b)$ or $(\bot, 0)$, while Alice learns nothing about the transmission.

When Bob receives $(\top, b)$, we call him *sapient*, and on $(\bot, 0)$ we say he is *ignorant*.

In *One of two OT*, denoted $\frac{1}{2}\mathtt{OT}(m_0, m_1)$, Alice posts two messages m_0 and m_1, and Bob receives either $(0, m_0)$ or $(1, m_1)$ with equal probability, while Alice learns nothing about the transmission. In *Chosen one of two OT*, denoted $\binom{2}{1}\mathtt{OT}(m_0, m_1; c)$, Alice posts two messages m_0 and m_1, and Bob provides a choice c, obtaining m_c while Alice learns nothing about the transmission. (These protocols are generally reducible to one another, reversible and precomputable based on a rich line of research.)

3 Background: Key Exchange, Extension

We consider "honest but curious" adversaries. Players follow the protocols, but their views should reveal nothing about sensitive values (secret keys, unselected messages). While the protocols can be strengthened against active attacks using standard techniques, most of the efforts to do so detract from understanding the algorithms and are thus left out of scope of this presentation.

3.1 Card Theory and Key Exchange

We consider only the basics of card theory, employing a simple deck of n unique cards, where for most of this exposition, we take the standard French suited deck with $n = 52$ or $n = 54$. Even the suits are "syntactic sugar" conveniences for the integer labels 0..51, although they will often be helpful for describing otherwise dry integer protocols.

We avoid specialized or duplicate cards. We restrict our attention to full factorial shuffles producing a permuted deck uniformly at random. In contrast, general Card Theory considers duplicate cards (eg den Boer AND, Crepeau Kilian Solitary), uniform cuts without shuffles, and even specialized cards of unusual shapes or partial disclosure.

In the 52-deck world, a line of classic results established card-based Secret Key Exchange (SKE) [13,14,35]. The earliest versions included a $(2, 1; 4)$ solution (Alice is dealt 2, Bob is dealt 1), which was proved minimally necessary to obtain guaranteed key establishment in any given deal [13].

In our setting, we relax the strict guarantees (and circumvent related lower bounds) in favor of obtaining success rates with reasonable expectation. This approach is more akin to Shannon Capacity, where any individual channel invocation (resp. deal) might be inadequate but over time the cumulative transmission rate is predictable.

Let us consider a simplified 3-card version of the 4-card SKE in [13]. The ideas are the same: establish a "key set," which is a set of two card values x and y, one held by Alice and one held by Bob. Eve gets to see the remaining card, yet if she learns only what the key set is, she still fails to know who holds x and y. Thus, a typical statement like "The key is 0 if Alice holds x, else it is 1" is enough to establish a one-bit key.

Let $\binom{S}{2}$ denote choosing a 2-set from set S, and let $\binom{S}{2|x}$ denote choosing a 2-set containing x from set S. Let $\delta(X)$ (or "is(X)") denote the binary truth value of whether statement "X" holds.

Dealt a hand $A = \{a\}$, Alice will choose a random 2-set $\beta = \{x, y\}$ containing a and another card. When Bob happens to hold one card in β, β is a key set and they've established a key. Half the time, Bob doesn't hold a card in β, in which case they need to deal again.

(For the nuanced expert: this lack of a guaranteed success at every single deal allows us to bypass a "4-card lower bound" in [13]. Even were the bound to apply, their more-involved 4-card protocol would apply here and throughout this work, *mutatis mutandis*, with little impact other than to cause the reader some grief.)

1-1-1 CardSKE using $(1, 1, 1; 3)$ deal

Alice on hand $A = \{a\}$		**Bob** on hand $B = \{b\}$
$\beta \leftarrow_\$ \binom{S}{2\|a} = \{\{a, a+1\}, \{a, a+2\}\}$	$\xrightarrow{\beta}$	
	$\xleftarrow{\texttt{acc}}$	$\texttt{acc} \leftarrow \text{is}(b \in \beta)$
	both abort if not $\texttt{acc}$	
$\texttt{key} \leftarrow \text{is}(a \in A)$		$\texttt{key} \leftarrow \text{is}(a \in A)$

Define the *capacity* or *rate* of a key exchange protocol to be the limit of $k(n)/n$ where $k(n)$ is the expected number of key bits established over n iterations.

Theorem 2. *(Winkler, Paterson, Rackoff, Fischer, Wright et al) 1-1-1 CardSKE finishes with probability* $1/2$ *and when it does, it establishes a secret bit between Alice and Bob. It has a secret key rate of* 0.5.

Proof. With six possible deals, there are six accepts and six rejects, and simple inspection shows the equiprobability of the established keys given Eve's view of the third card. Intuitively, Eve cannot distinguish (a, b) from (b, a) and therefore learns nothing about `key`.

3.2 Extension

Pseudorandom number generators extend short random strings to long ones. With a short secret key and a one-way function, parties can extend their key material to polynomially-larger strings.

The same is true of OT: a one-way function suffices to extend a handful of base OT's to polynomially many more, as first shown in [4] and later strengthened with an elegant and simple algorithm in [19]. Without diving into excessive detail, Alice can request that Bob hand her λ bits obliviously by way of (reversed) base OTs. She obtains a λ-bit string s which they can then repeatedly apply to help hide one or the other of a larger, polynomial number of new messages. For concreteness, the intuition has a flavor of encrypting $E(m_0)$ and $E(m_1 \oplus s)$ sometimes, vs $E(m_0 \oplus s)$ and $E(m_1)$ at other times. For this paper's scope, we do not need details but we aim to claim convincingly that Alice and Bob can feed their card-based base OTs into one-way-function driven programs that will handle the higher layers of repeated Mental Poker.

Our pragmatic goal is to produce $\lambda = 128$ OTs from Five-Card Draw, if possible, especially if it can be done affordably.

4 Trios and Packing

The *One if by Land, Two if by Sea* flavor of card keyset SKE suggests that a two-card deck could implement OT. Deal $(0, 1; 2)$, namely a single card from

$\{Ah, As\}$ to Bob, leaving one in the draw pile. Alice might say, "$m = 1$ if you got Ace of Hearts, $m = 0$ if you got Ace of Spades." Problematically, Alice doesn't know which card Bob received, namely she doesn't know a keyset and can't formulate an effective transmission. If we permitted her to have a keyset (e.g. deal $(1, 1; 2)$), she would know everything.

The keyset (a, b) in a 1-1-1-SKE deal is notable in that it materializes a flaky secret channel. The failures are useful. Alice reaches out to connect a keyset to Bob, but half the time Alice connects her keyset to Eve.

This pattern seems applicable to Rabin OT(m), even though Eve isn't around. If Alice can connect a keyset to Bob or miswire it away from Bob, without knowing where the far end of the keyset goes, then she can obliviously transmit on the keyset. Obviously, we must be sure that when Bob lacks internal details of the keyset, he learns nothing; there are pitfalls.

We pursue this approach, strengthening it to obtain $\binom{2}{1}\mathtt{OT}$, and then see how to harvest trios from $(5, 5; 52)$ Mental Poker deals.

4.1 Trios for Rabin OT

Deal a card to each player, leaving one in the draw pile. A naive *OneIf* keyset labeling is dangerous, however. If Alice deterministically announces "$m = 1$ if I hold a" when dealt a, Bob learns m immediately from inspecting her implicit admission to holding a. She is naively mentioning a only when she has it.

There are a few ways to fix this situation. One is to admit defeat and justify it. Having admitted defeat, one can pursue alternative physical ideas, like writing data on the fronts of cards and enjoying a markedly artificial scenario where Bob receives one string (somewhat begging the overall question). This approach turns out to inspire other productive avenues, however, so we will later illuminate some of them. There are reasons to justify defeat, too, such as the limitations of OT extractability from Generalized Erasure Channels, and those are also illuminating.

For now, let's fix with randomization. Alice picks a random bit q, which can later be used as a pad with an intended message m. Alice will send one of a couple of messages (based on a coin flip ρ), either of which will enable a sapient (keyset-card holding) Bob to infer q. The messages are equally likely when viewed by an agnostic Bob, hiding q. Let $\hat{a}$ denote the companion member of a in the two-element set β.

1-1-1 CardRabinOT using $(1,1;3)$ deal

Alice on hand $A = \{a\}$ **Bob** on hand $B = \{b\}$

$\beta \leftarrow\!\!\$ \binom{S}{2|a}$ // $\beta = \{a, \hat{a}\}$

$q, \rho \leftarrow\!\!\$ \{0, 1\}$

$\texttt{msg} \leftarrow$ "use q if is$(a \in A)$" if $\rho = 0$

else "use $\bar{q}$ if is$(\hat{a} \notin A)$"

$\xrightarrow{\beta, \texttt{msg}}$

$(\top, q)$ if $b \in \beta$ else $(\bot, 0)$

Theorem 3. *1-1-1 CardRabinOT implements Rabin OT on a random message q with rate* 1.

Proof. For sapient Bob, inspection shows how he calculates q correctly in all cases, knowing how to interpret `msg`. An agnostic Bob may encounter messages among six enumerable cases (three for a, two for q), of which four have nonzero probability in a given run (having ruled out b). Of those four, two correspond (with equal weight) to $q = 0$, and two correspond to $q = 1$. The two corresponding to $q = 0$ have equal probability given $q = 0$ over runs where Alice received a and runs where Alice received $\hat{a}$, thus β provides no information to Bob. Thus, Bob's view is independent of β, a, q given b.

4.2 Trio Pairs for Chosen OT

Ordinarily, one can summon OT reductions from the literature to implement $\frac{1}{2}\texttt{OT}$ from `rabOT`, although often quite costly. In our overarching goal of supporting pragmatic Mental Poker from Poker, excessive invocations could be such a drain that we'd turn to other avenues like the aforementioned card marking and soon-to-visit deck building. But in the honest-but-curious setting, we can obtain good performance by having Bob pair up sapient and agnostic executions.

Keep in mind the notion of a keyset as a wire which goes to Bob or goes to ground. If we pair one invocation where the wire goes to Bob and another invocation where the wire goes to ground, we have a model for obtaining $\frac{1}{2}\texttt{OT}$. Alice applies m_0 to the first `rabOT` and m_1 to the second; only one wire transmits it.

Let us stretch our notation so that $\{<\}\, v, b >$ denotes the full data surrounding a Rabin OT (in contrast to how we have annotated $(v, \cdot)$ for what Bob receives, which can omit the input.)

2x3-card $\frac{1}{2}\texttt{OT}(m_0, m_1)$

$\{v_0, q_0\} \leftarrow \texttt{3CardRabOT}()$

$\{v_1, q_1\} \leftarrow \texttt{3CardRabOT}()$

abort if $\{v_0, v_1\} \neq \{\top, \bot\}$

$\mathcal{A} \rightarrow \mathcal{B} : (q_0 \oplus m_0, q_1 \oplus m_1)$

Clearly, `2x3Card12OT` succeeds half the time, giving it a rate of 0.5. Getting a rate approaching 1.0 is straightforward by amortizing over several iterations of the 3-card protocol:

kx3-card $\frac{1}{2}\texttt{OT}(\{(m_{i0}, m_{i1})\})$

repeat $20k$ times : $\{v_i, q_i\} \leftarrow \texttt{3CardRabOT}()$

$\texttt{numsucc} \leftarrow |\{v_i : v_i = \top\}|$

abort if not $9k \leq \texttt{numsucc} \leq 11k$

Bob announces k pairs of $(\top, \bot)$ instances

Proceed as in 2x3-card $\dfrac{1}{2}\texttt{OT}$

4.3 Packing and Harvesting in Larger Decks

We set out to address 52-card Poker – and 52-card Poker shall we address.

The theoretical and practical efficiency of our goals are affected by adhering to the hand we are dealt, so to speak. Here, we show that we can harvest trios with reasonable efficiency from poker deals.

In the preceding section, we explored using multiple independent invocations of "special decks" (3-cards). Let's now adapt those ideas to Five Card Draw.

If we try to pack 17 copies of a 3-subdeck into a 52-card deck, we have immediate blockers. Obviously, each 3-subdeck interferes with the others - sometimes Alice or Bob get no cards, sometimes multiple cards, from any particular 3-subdeck. The annoying 52^{nd} card can get in the way.

Rejection sampling comes to the rescue. Alice and Bob will each announce their opinion of the viability of each 3-subdeck. Where they both approve, that subdeck will be used to implement a desired 3-card invocation.

It is evident that there are highly conditional dependencies on the disruption of one subdeck and another (*e.g.* when Alice receives all 3 cards of subdeck i, her chances of receiving a single card in deck j are diminished). Fortunately, for any subdeck j, conditioned on Alice receiving one card, Bob receiving one, and one remaining in the draw, the subdeck j behaves like an independent 3-deal.

The success rate of packing trios into a 52-card deck is purely combinatorial and can be observed (Table 1) with high assurance using Monte Carlo simulations. Pleasantly, the maxima are reached when each player receives 1/3 of the deck, although to stay within Poker models, Alice and Bob can at best reach 10 cards each by discarding their initially dealt 5-card hands to draw 5 more.

Table 1. Trio Success Rates in 52-card Shuffles

Hand A	Hand B	Trio Rate	Remarks
5	**5**	**0.808**	initial deal
10	**10**	**2.462**	each draws 5
17	17	*4.004*	not poker
18	18	3.979	not poker
15	10	3.118	not poker

For n-sized decks (multiple of 3), the closed-form maximal success rate is given by a colored ball without-replacement task using $n/3$ colors, asymptotically $\frac{2}{27}n$.

$$\frac{2n^3}{27(n^2-3n+2)}$$

Pleasantly, $\frac{2}{27}51 = 3.778 \approxeq 4.004$ per simulation. Again: this case is for dealing optimally large hands from uniform shuffles, not proper 5-Card Draw.

Most importantly, we can expect to get 2.462 Trios per 52-shuffle and can build base $\binom{2}{1}\mathtt{OT}$ from there with quite reasonable parameters:

Claim. 5-Card Draw with full shuffles supports a rate of $1.231\binom{2}{1}\mathtt{OT}$, namely $128\binom{2}{1}\mathtt{OT}$ per 104 52-shuffles. Establishing 128 $\binom{2}{1}\mathtt{OT}$ costs about a hundred games.

5 Marked and One-Time Decks

This section is largely "card cryptography," as we'll discuss unuusal manipulations of cards, staying nevertheless within the bounds of "French 52-deck, shuffling machine, dealer."

The headwinds of low data rates and threatening impossibility concerns inspire alternative thinking. Looking ahead, there are GEC-related works which could suggest OT is impossible. (The intuition there is that a shuffle is independent of an input, and regarded as a channel without side communication, this suggests asymmetric knowledge could be unreachable.)

We change the rules briefly, allowing Alice to write on the faces of her cards, to explore what we can achieve. At the very least, it seems she can write large messages and obtain rich string-based OT.

This does cost us a deck, since re-using a marked deck would compromise previous interactions. Some scribbling for erasing is conceivable, and there are pragmatic crosshatching tricks for arthritic scribes ($-$ for 0, $|$ for 1; recycle by overstriking all to $+$), but still: a finite limit. "Finite-time decks."

5.1 Unary

A crude but easy approach is to implement `rabOT` by writing a random string s redundantly on several cards. If Alice repeats s on 6 cards, then the probability that Bob receives at least one marked card in a $(5,5;52)$ deal is 0.52. With 7 repetitions, the probability goes to 0.48. To build an effective OT stream, one wants the agnostic cases to be at least half, thus 6 cards is optimal for a Unary Marking.

Using the full $(10,10;52)$ power of drawing extra cards does not help. With 3 cards marked with s, Bob learns s with probability 0.52 as before. With 4 cards, his discovery rate goes to 0.44.

The Unary Marking is simple and effective, but it produces `rabOT` at roughly rate 1 per deal. With a need for 128 chosen OTs, one can expect about 250 hands to bootstrap Mental Poker.

5.2 Trio

Let's revisit the Trio. Alice writes a random string on every card (in the Trio; ultimately, on every card in a 52-deck). Bob is dealt one of the three (b), learning one string (s), agnostic to the others.

Alice, having card a, knows that Bob has one card in $S - \{a\}$. She randomly names the two cards as $\{x_0, x_1\}$ (their scribbled strings being $\{s_0, s_1\}$) and announces that she will relay $(m_0 \oplus s_0, m_1 \oplus s_1)$.

Clearly, Bob will learn m_i for the i such that $b = x_i$. This gives $\frac{1}{2}\mathtt{OT}(m_0, m_1)$. Via standard methods, if we like, Bob can ask Alice to reverse her choice to $(m_0 \oplus s_1, m_1 \oplus s_0)$, obtaining $\binom{2}{1}\mathtt{OT}(m_0, m_1; c)$.

Claim. 5-Card Draw with full shuffles supports a rate of $2.462\binom{2}{1}\mathtt{OT}$ using marked Trios. Establishing 128 $\binom{2}{1}\mathtt{OT}$ costs at most about fifty games.

5.3 Doubles

When discussing marked decks, there has been a small sleight of hand in play. The Trio approach is excessive. Alice need only mark two cards to convey the two bits she's sending. This scenario is basically definitional: Bob receives one of two message-bearing cards as an implementation of $\frac{1}{2}\mathtt{OT}$. Marking is powerful.

We did set out to conquer poker deals, however. Returning to packing: 26 doubles into a deck. The $i^t h$ pair of random strings is written on cards $(2i, 2i+1)$. Bob reports to Alice the list `viable` of i such that he received exactly one card in $\{2i, 2i+1\}$. For each $i \in$ `viable`, Alice uses the markings (s_{2i}, s_{2i+1}) to convey a pair of messages as $(m_0 + s_{2i}, m_1 + s_{2i+1})$.

The combinatorial analysis considers ten balls selected without replacement from a bag of 52 balls, two of each color, seeking the expected number of colors where Bob picked exactly one. This gives

$$26 \cdot \frac{\binom{2}{1}\binom{50}{9}}{\binom{52}{10}} = 26 \cdot \frac{70}{221} \approxeq 8.235.$$

Claim. 5-Card Draw with full shuffles supports a rate of $8.235\binom{2}{1}\mathtt{OT}$ using marked Doubles. Establishing 128 $\binom{2}{1}\mathtt{OT}$ costs about sixteen hands.

52 Cards but Not Quite Poker. Dealing $(0, 26; 52)$ goes a long way. It's not a Poker hand but it illustrates the power at hand. The viability rate is a combinatorial exercise: sample $n/2$ balls without replacement from n balls enjoying $n/2$ colors and 2 balls of each color. The expectation is

$$\frac{n^2}{4(n-1)}$$

which for $n = 52$ is 13.25.

Claim. Dealing $(0, 26; 52)$ with full shuffles supports a rate of $13.25\binom{2}{1}\mathtt{OT}$ using marked Doubles. Establishing 128 $\binom{2}{1}\mathtt{OT}$ costs about ten deals.

5.4 Blending Marking and Building

Let's buy two decks. One, keep untouched. The other, put a dot on every face and then throw away the pen. The backs are all the same.

Instead of marking a random one-bit number on 52-cards as in the previous marked-deck protocol, Alice *builds* a 52-deck using the 104 cards at hand. She picks a random Ah from the dotted or undotted deck. She picks a random $2h$. And so on, until she has a deck whose cards are the standard 0..51 but randomly marked with 0 (no dot) and 1 (a dot). A deck might look as follows:

$$Ah, 1h, 2\dot{h}, 3h, 4\dot{h}, 5\dot{h}, \ldots, Qc, K\dot{c}$$

representing markings

$$001011\ldots01$$

It's not hard to see that Alice and Bob can implement a Marked Deck protocol this way. But now, they've got unlimited use. Alice can rebuild a fresh "Marked Deck" for every deal without having to mark anything anew. She might need to flip coins on her own (or use the shuffling machine thoroughly and take the first candidate from any dot/no-dot pair). We can invoke Claim 5.3 to eke out $8.235\binom{2}{1}\mathtt{OT}$ per game, and in other words, *sixteen hands of poker with a deck built from two decks is enough to bootstrap secure computation.*

"Poker Face," indeed.

6 Deck Building

There are many entertaining games (like Magic the Gathering) where selecting a starting deck from a library of cards is part of the gameplay and strategy. Here, we turn to deck building in the novel pursuit of *game implementation.*

Let $(h_0, h_1; n/m)$ denote dealing hands of size h_0 and h_1 from an n-card deck built as a subset of an m-card collection. For simplicity at present, we consider all m cards to be unique.

We saw in §5 that building a $n = 52$ deck from a collection with $m = 104$ cards can provide very high OT rates. Using claim 5.2, an expected 16 chosen OTs can be harvested from one Five-Card Draw deal, $(0, 10; 52/104)$.

We note again that we first toyed with permitting Alice to ignore her hand. If Alice sees her hand, which is reasonable, she destroys the Doubles that she sees, changing the overall rate to $4.320\frac{1}{2}\mathtt{OT}$ per $(5, 10; 52/104)$ deal.

The power of building is evident: these rates are higher than the engineered Trio approach inspired by Card SKE. This first setting costs two decks, but the building process is simple (picking dotted/undotted at random) and the interpretation steps are simpler.

6.1 Joker's Wild

Start with a single 54-card commercial deck, with a red and black joker (Jh, Js). Pro forma: these are cards 52 and 53. We consider Five-Card Draw on a built deck with deal signature $(5, 10; 52/54)$.

Represent bits $\{0, 1\}$ as $\{As, Ah\}$, namely: if Bob receives the Ace of Spades, he concludes $m = 0$, if Bob receives Ace of Hearts, he concludes $m = 1$, and if he receives neither (or both) he concludes $\perp$. Naturally, this setup as-is won't transmit anything (it is a GEC with no input message dependence 8).

Alice takes action. On input $m = 0$, Alice replaces Ah with Jh. Bot may receive As in his hand and conclude $m = 0$, else he will conclude $\perp$, but he will never conclude $m = 1$. Likewise: on input $m = 1$, Alice replaces As with Jh. Similarly, Bob can end with $m = 1$ or $\perp$ but not $m = 0$.

There's a $10/52$ chance that Bob receives the Ace that will inform him of m. This extremely simple deck-building puts us within well-studied results on OT with altered transmission rates. It's a GEC with erasure probability $p = 42/52$, if Alice avoids seeing her own cards. (If she is forced to look, she may receive the encoding Ace and understand that Bob got $\perp$. There are protocols in the literature to manage these kinds of OT variations.)

Alice can redundantly code m with the second Joker Js replacing the undesired partner in, say, $\{Ac, Ad\}$. Bob will receive m with roughly $20/52$ probability and remain agnostic with roughly $p = 32/52$ probability. In other words, the redundant version achieves roughly 60% erasure - close to one Rabin OT per shuffle. (Using the honest-but-curious pairings approach, a rate of roughly 0.4 $\binom{2}{1}\mathtt{OT}$ per hand is feasible.)

6.2 Double Capacity Jokers and General Parameters

There are many directions in this design space. The two Jokers can be applied toward two different message bits, for example. Naturally, they interfere somewhat.

But Jokers are just two cards out of 52. There's nothing magical or distinctive about them. There are $\binom{54}{52} = 1431$ different starting decks to build and deploy, suggesting quite a few more opportunities to extract better results.

The trade-offs in OT capacity for different build strategies are outside the scope of this paper but are under investigation and a fruitful direction for more general research. Investigating Mental Poker as a baseline has opened up a new avenue in card cryptography: n-deck building from m cards.

7 Deck Building via Rejection Sampling

Our final foray into poker-generated OT returns to the original setting of a pristine 52-card deck. No markings, no building. The ideas in the marking and building investigations inspire us, however.

In particular, recall the Joker's Wild protocol of §6.1. Coding $\{0, 1\}$ with cards $\{As, Ah\}$, Alice removes the unwanted Ace by replacing with a joker.

Alice can achieve the same effect using rejection sampling. Simply execute a $(5, 5; 52)$ deal (or respectively $(10, 10; 52)$ where the parties draw their allotted extra cards). If Alice draws the unwanted Ace, she permits the protocol to continue. If not, she tells Bob to abort using this hand.

Concretely, say $m = 0$. Then Alice wishes to ensure Bob never sees Ah but has some $10/52$ chance to see As and infer $m = 0$. In Joker's Wild, Alice replaced Ah with a Joker to obtain a 52-card deck to feed into the shuffler.

Now, instead, in the rejection sampling approach, Alice waits until she holds the Ah. She knows for sure that Bob doesn't have her. Have it. I mean.

Anyway, life is sorted when Alice holds the hearts.

Conditioned on Alice holding the unwanted Ace, this Rejection protocol produces OT at the Joker's Wild rate. To obtain this condition costs a factor of roughly $52/10$ times as many deals. In other words, the OT rate drops by a factor of about 5.2 compared to Deck Building with signature $(10, 10; 52/54)$.

8 Limits of the Full Factorial Shuffle

There are many headwinds pushing back on full-shuffle deck-based protocols. These range from characterizations of transmissions as non-interactive channels, to one-party game settings (hiding information from oneself to play solitaire), and the observation that the literature of card-based calculations utilizes cuts (rotations) rather than full shuffles (which destroy most or all proposed protocols).

8.1 den Boer and other AND protocols

To inform this presentation, we make the rotation regime concrete by recalling den Boer's celebrated Five Card Trick [6] to compute and reveal AND(x,y).

Denote a face-down card using brackets $[z]$. Den Boer's trick uses three indistinguishable hearts and two indistinguishable spades. Alice controls one heart and one spade. Bob controls one heart and one spade.

At the start, one heart is placed face down in the middle:

$$* * [h] * *$$

Mnemonically, the overall result will be 1 when there are three hearts together, but it will be 0 when hearts are divided.

Alice places her cards to the left. On input $x = 1$, she places her heart next to the center heart. Otherwise, on $x = 0$, she separates the hearts.

$$x = 0 : \quad [h][s][h] * * \quad \text{or} \quad x = 1 : \quad [s][h][h] * * \tag{1}$$

Bob places cards similarly on the right: if $y = 0$, he keeps hearts together, else he divides them.

Table 2. den Boer Five Card AND Trick [6]

AND	$y = 0$	$y = 1$
$x = 0$	[h][s][h][s][h]	[h][s][h][h][s]
$x = 1$	[s][h][h][s][h]	[s][h][h][h][s]

Starting with the cases in Table 2, the five face-down cards are rotated randomly - Alice and Bob apply random cuts. Finally, all are turned up.

Notably, on $(0, 0)$, there are always three hearts together (viewed circularly). The other three input representations are already circular rotations of each other, thus any random spin will lead to the same uniform distribution on *hshsh* rotations, all of which separate some heart and thus represent $AND = 0$.

Like large swaths of protocols [17,21,23,25,27,29], den Boer and others make use of a uniform cut (rotation). Clearly, a full shuffle would completely damage the needed invariant that three hearts are adjacent iff the result is 1.

8.2 Solitary Games

A very creative use of card crypto is the Solitary Game, first approached by Crepeau and Kilian [10]. In a manner similar to den Boer's trick, they show how to achieve solo games - where there is information hidden from the solo player - using rather long sequences of repeated cards with cuts as a fundamental randomizer. The protocol shows theoretical feasibility but is impractical.

Full shuffles would destroy their proposed methods. They double down [10, p. 329]:

Conjecture 4. (Solitary Games Conjecture) Full shuffles do not suffice to implement solitary games.

We have seen in marking and building decks that it is essential to introduce some symmetry breaking in the form of having the deck depend in some way on the inputs at hand. The mechanisms of talking, writing or rejecting generally presuppose some asymmetric action and communication. Alice rejects a current deal based on secret knowledge, Alice validates a Trio by stating a summary of secret knowledge, Alice writes secret values on cards, Alice builds a deck based on secret substitutions. In solitary games, the player has no counterparty (except self) to keep secrets from, and communications are essentially from self to self anyway, with no distinction between "sender" and "receiver." The Solitary setting grants no effective powers to Alice to break symmetry as is done generally elsewhere, including in this work, and thus it is unsurprising that when alone, she would be constrained to avoid completely symmetric full shuffles.

8.3 Generalized Erasure Channels

OT can be described as a noisy channel, erasing its input half the time. The Generalized Erasure Channel [1, 18, 28] broadens this view to any discrete memoryless channel which occasionally erases its input.

Definition 5. *A* Generalized Erasure Channel *is a discrete, memoryless channel $W : X \to Y$ where Y is partitioned into $Y^{\top} \cup Y_{\perp}$ such that $W(y|x) = W(y)$ for any $y \in Y_{\perp}$.*

Definition 6. *For a GEC W, define the erasure rate $p^{\perp} = \sum_{y \in Y^{\perp}} W(y)$, and define $W_{\top}(y|x) = \frac{1}{1-p^{\perp}} W(y|x)$ for $x \in X, y \in Y_{\top}$.*

Extracting string OT for long strings from GECs is a natural task. The following is shown in [1, 18, 28] for various adversarial models including actively malicious ones:

Theorem 7. *For a GEC with $p_{\perp} \geq 1/2$, the OT capacity is $(1 - p_{\perp})C(W_{\top})$, where $C(W_{\top})$ is the Shannon capacity of the discrete memoryless channel $\{W_{\top} : X \to Y_{\top}\}$.*

An essential characteristic to note is the need for some dependence on inputs x. A full shuffle provides the "receiver," Bob, with a hand y of cards. The hand Bob receives is not reliant on any input provided by Alice. Intuitively, the shuffle as a strict channel does not transmit any helpful information: $p_{\perp} = 1$, and the resulting OT capacity is null.

The important waiver, however, is the absence of communication. When sender and receiver can communicate, they can take any particular GEC and use rejection sampling to obtain a derived GEC which does, in fact, behave conditionally differently for different inputs.

This is another aspect in which full shuffles seem fruitless in settings like Solitary Games. The solo player is faced with trying to modify a wholly independently random primitive using communications to herself, but those communications beg the question of asymmetric knowledge. Were the Solitary Games Conjecture to fail, the solitary games built on full shuffles would be tools to implement OT using self-to-self input-independent channels, implying OT implementable from scratch, which is not information-theoretically possible.

9 Conclusion

This work has developed a paradigm to build and extend Mental Poker from Poker in a structured fashion:

1. Obtain base OT via 5-Card Draw Poker on fully shuffled decks (information theoretic)
2. Extend OT cryptographically via one-way function [4, 19] (computational)
3. Apply 5-Card Draw circuit on extended OT (info. theor. or computational)

Some steps enjoy information-theoretic strength, such as the extraction of OT from fully-shuffled decks.

We shed light on challenges in some domains where full shuffles appear detrimentally over-randomizing, such as solitaire card games or non-interactive erasure channels where "conversations" are lacking. Strikingly, those settings led to impossibility conjectures whose nature is now clarified. With feedback communication and rejection sampling, this work shows a way to extract generalized erasure channels from fully-randomizing full-permutation settings. Solitaire and one-way channels fail to benefit because they can't support the collaborative communication involved in OT extraction.

In the realm of card cryptography, we've introduced and investigated the novel approaches of marking and building decks, with concrete improvements in OT extraction rates. The one-time deck and the finite-time deck are interesting objects of study, and we showed how building n-card decks from $m = 2n$ card collections is a marking-free way to emulate certain marked decks and obtain high-rate OT. *Game implementation theory* using deck building arises as a novel application of deck building, otherwise ordinarily motivated for strategic gameplay.

Our protocols are *Paris algorithms*, as opposed to "Monte Carlo" or "Vegas": having gathered in Paris to play with cards, Alice and Bob can then play remotely and indefinitely. Depending on the options they choose - complexity of the encodings, marking or building of decks - they can set up 128 $\binom{2}{1}\texttt{OT}$'s in 16-100 hands of 5-Card Draw with full deck shuffles. An afternoon in Paris or a night in Casablanca is enough.

Thus we connect the ouroborous: mental poker can be based on and extended indefinitely from poker, real or mental.

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